

Acupuncture Research

Transcutaneous Electrical Acupoint Stimulation Improves Postoperative Cognitive Function in Senior Patients Undergoing Video-Assisted Thoracoscopic Surgery: A Randomized Controlled Trial*

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ABSTRACT **Objective:** To evaluate the effectiveness and safety of transcutaneous electrical acupoint stimulation (TEAS) for improving postoperative cognitive function in senior patients undergoing video-assisted thoracoscopic surgical (VATS). **Methods:** From January to December 2020, 97 participants were randomly assigned to the TEAS group (49 cases) and the control group (48 cases) by a random number table. The patients in the TEAS group received TEAS, at the bilateral Neiguan (PC 6) and Zusanli (ST 36) acupoints. The control group received sham TEAS. The stimulation was started from 30 min before surgery until the end of the operation. The primary outcome was the incidence of postoperative cognitive dysfunction (POCD), diagnosed based on the changes in the Mini-Mental Status Examination (MMSE) and Montreal Cognitive Assessment (MoCA) scores. The secondary outcomes were plasma levels of S100 β protein and neuron-specific enolase (NSE). **Results:** The incidence of POCD on day 1 and 3 after surgery in the TEAS group was significantly lower than that in the control group [day 1 after surgery: 28.3% (13/46) vs. 52.3% (23/44), $P=0.028$; day 3 after surgery: 21.7% (10/46) vs. 40.9% (18/44), $P=0.043$]. Compared with baseline, the MMSE and MoCA scores decreased to various extents in both groups. The MMSE scores on day 1, 3, and 5 after surgery and MoCA scores on day 1, 3, 5, and 7 after surgery in the TEAS group were higher than those in the control group (all $P<0.05$) in both groups. Compared with baseline, the plasma levels of S100 β and NSE were significantly increased at 4, 8, 12, 24 h after surgery (all $P<0.05$). Compared with the control group, the plasma levels of S100 β and NSE were lower in the TEAS group at 4, 8, 12, and 24 h after surgery (all $P<0.05$). No obvious adverse events were found during the trial. **Conclusion:** Application of TEAS in senior patients after VATS could reduce incidence of POCD and improve postoperative cognitive function.

KEYWORDS transcutaneous electrical acupoint stimulation, video-assisted thoracic surgery, senior, postoperative cognitive dysfunction

Postoperative cognitive dysfunction (POCD), a severe postoperative neurological sequela, is mainly characterized by dysfunction related to cognition, orientation, thinking, and memory.^(1,2) The incidence of POCD in the early postoperative period is reportedly higher in senior patients due to organ dysfunction, central nervous system degeneration, and decreased tolerance to posttraumatic stress.⁽³⁾ POCD is generally transient, but it can progress to permanent cognitive decline and increased overall mortality.^(4,5) POCD can prolong the length of hospital stay, increase treatment cost, and improve the patient's quality of life, all of which can result in poor work capacity.⁽⁶⁾

The pathogenesis of POCD remains unclear, and there is a lack of clinically effective treatment

methods. Recently, the application of physical methods to prevent the occurrence of POCD has become a research hotspot. A clinical trial has confirmed that electroacupuncture can inhibit the inflammatory response, manage the balance of pro-inflammatory and anti-inflammatory factors, exerts obvious effects on brain

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protection, and can effectively prevent or reduce the occurrence of postoperative POCD in senior patients.⁽⁷⁾

Transcutaneous electrical acupoint stimulation (TEAS) combining transcutaneous electrical nerve stimulation and acupoint therapy is a noninvasive treatment option that is easily accepted by patients. As a result, TEAS has gradually replaced acupuncture in clinical practice. Current scientific research data support that TEAS preconditioning could attenuate brain injury by inhibiting the oxidative stress response, so as to protect cognitive function during perioperative period.⁽⁸⁾ However, the effect of TEAS on postoperative cognitive function in elderly patients with lung cancer remains unclear. The purpose of this study was to evaluate the effectiveness of TEAS on postoperative cognitive function in the elderly patients with lung cancer, and the results may provide a new direction for the prevention of POCD in the elderly.

METHODS

Inclusion and Exclusion Criteria

All participants met the following inclusion criteria: (1) age >60 years; (2) American Society of Anesthesiologists Physical Status Grades I–III; (3) undergoing video-assisted thoracoscopic surgical (VATS) lobectomy under general anesthesia. The exclusion criteria were as follows: (1) participants with impaired cognitive function or with abnormal mini-mental status examination (MMSE, <24) or Montreal Cognitive Assessment (MoCA, <26) scores^(9,10) before surgery; (2) participants with obvious functional abnormalities of the heart, lung, liver, or kidney; (3) a history of traumatic brain injury, neurological diseases, and alcohol abuse; and (4) participants who were unable to communicate with doctors effectively and cooperate with them to complete the cognitive function test.

Patients and Blinding

From January to December 2020, 97 participants were included from North China University of Science and Technology Affiliated People's Hospital and randomly assigned to the TEAS group (49 cases) and the control group (48 cases) by a random number table. This study adopted a single blind design, and only acupuncture operators know the grouping of patients. A sealed envelope containing an allocation sequence number for each participant was opened after each participant met eligibility criteria and informed consent was obtained. The study was approved by the Ethics Committee of Tangshan People's Hospital

(No. RMY-LLKS-2020-044) and followed the declaration of Helsinki. The information of the study was explained to all enrolled participants, and written informed consent was obtained from each participant.

Sample Size Estimation

According to the previous study,⁽⁹⁾ the values of MoCA score in 9 d post-treatment in patients after transurethral resection of prostate were 25.4 ± 1.3 in the experiment group and 23.8 ± 2.4 in the control group, respectively. With non-inferiority test, inspection level $\alpha=0.05$, grasp $1-\beta=0.90$, the sample size should be 31 cases in each group. Considering the 20% lost follow-up rate, at least 78 samples should be included in the study, and 90 patients were selected for the study.

Intervention

TEAS was conducted by an experienced acupuncturist. The bilateral Neiguan (PC 6) and Zusanli (ST 36) points were selected as the acupuncture points based on the Chinese medical theory. Electrical stimulation was performed using the Hwato electronic acupuncture treatment instrument (Model SDZ-V, Suzhou Medical Appliances Co., Ltd., Suzhou, China). Electrode tabs were placed on bilateral PC 6 and ST 36 points after skin disinfection.

The TEAS group received acupoint electrical stimulation at 30 min before the induction of anesthesia. TEAS was performed at an alternative dense-disperse frequency of 2/10 Hz (2 Hz for 10 s and 10 Hz for 5 s). The optimal intensity ranged from 6 to 15 mA, it was regulated to maintain a slight twitching of the regional muscles based on individual maximum tolerance. The stimulus was maintained until the end of surgery. The control group had electrode pads applied but received no electrical stimulation.

Primary Outcomes

The MMSE and MOCA scores were used to evaluate the cognitive function on day 1 before surgery (baseline) and on day 1, 3, 5, and 7 after surgery by the same medical staff. The MMSE scale was used to grade 7 dimensions: orientation, language, recall, attention, execution, spatial capability, and calculating ability. The total score was 30 points, with a score of 24–27 points suggesting slight cognitive impairment (CI), 19–23 points suggesting moderate CI, and 0–18 points suggesting severe CI. The MoCA scale was used to evaluate the following 8 dimensions (total score 30 points): spatial

and executive capability, memory, attention, naming, language, delayed recall, abstract thinking, and orientation. If the subject had ≤ 12 years of education, 1 point was added to their score. A score of < 26 indicated impaired cognitive function. POCD was diagnosed when postoperative MMSE and MoCA scores were declined by > 1 standard deviation, when considering the baseline value.

Secondary Outcomes

Internal jugular venous blood sample (3 mL) was drawn at 30 min before induction of anesthesia (baseline) as well as 4, 8, 12, and 24 h after surgery. The blood samples were maintained at room temperature for 30 min and centrifuged at 3,000 r/min for 10 min before the upper serum layer was collected and stored at -80°C . The plasma levels of S100 β protein and neuron-specific enolase (NSE) were detected by enzyme-linked immunosorbent assay, strictly adhering to the manufacturer's instructions (Wuhan Boshide Biological Engineering Co. Ltd., China).

Safety

Safety evaluation was performed for all patients who participated in the randomization procedure during the whole study. Throughout the trial, adverse events caused by TEAS or medicine were observed and recorded in both groups, such as fainting, skin allergy or heart rate irregularities, etc.

Statistical Analysis

Statistical analysis was performed by using SPSS version 19.0 (SPSS Inc., Chicago, IL, USA). Continuous variables were expressed as mean $\pm$ standard deviation ($\bar{x} \pm s$) and processed with Student's *t* test. Categorical variables were expressed as numbers (percentage) and processed with Chi-square test. The MMSE and MoCA scores were compared, matched for age, gender, and formal education by using Z-score analysis. Data of MMSE and MoCA scores, levels of plasma S100 β protein and NSE were analyzed by two-way repeated-measures with Bonferroni *post hoc* comparisons. A value of $P < 0.05$ was considered statistically significant.

RESULTS

Trial Completion Condition

This trial recruited 106 patients, among whom 97 patients were eligible. However, only 90 participants (46 in the TEAS group and 44 in the control group) were enrolled in this study (Figure 1). The baseline

characteristics and demographic data were statistically insignificant between the two groups ($P > 0.05$, Table 1).

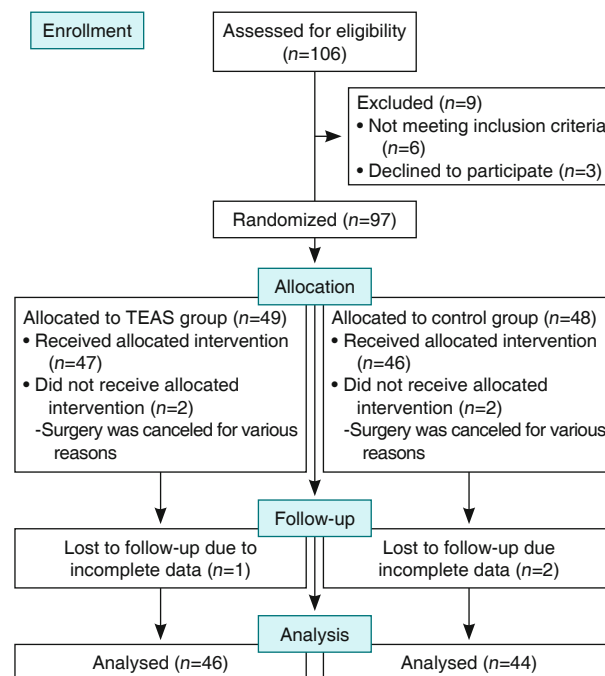


Figure 1. Flow Chart of Study on TEAS in Senior Patients Undergoing Video-Assisted Thoracoscopic Surgery

Note: TEAS: transcutaneous electrical acupoint stimulation

Table 1. Demographic and Surgical Characteristics of Patients ($\bar{x} \pm s$)

Characteristic	TEAS group (46 cases)	Control group (44 cases)
Age (Year)	72.32 $\pm$ 10.21	72.14 $\pm$ 11.34
Gender [Case (%)]		
Male	24 (52.2)	22 (50.0)
Female	22 (47.8)	22 (50.0)
BMI (kg/m ²)	23.63 $\pm$ 4.01	22.37 $\pm$ 5.74
Education status (Year)	10.22 $\pm$ 5.91	10.48 $\pm$ 6.25
History of alcohol intake	18 (39.1)	14 (31.8)
Hypertension	16 (34.8)	14 (31.8)
Diabetes	5 (10.9)	4 (9.1)
ASA rating		
I	32 (69.6)	33 (75.0)
II	14 (30.4)	11 (25.0)
Side of surgery		
Left	22 (47.8)	22 (50.0)
Right	24 (52.2)	22 (50.0)
Duration of surgery (min)	165.26 $\pm$ 17.47	168.48 $\pm$ 15.38
Anesthesia time (min)	187.47 $\pm$ 21.62	188.58 $\pm$ 25.27
Volume of blood loss (mL)	216.41 $\pm$ 42.29	221.79 $\pm$ 56.16

Notes: ASA: American Society of Anesthesiologists; BMI: body mass index

POCD Incidence and MMSE and MoCA Scores

The incidences of POCD on day 1 and 3 after surgery in the TEAS group were significantly lower than those in the control group [day 1 after surgery: 28.3% (13/46) vs. 52.3% (23/44), $P=0.028$; day 3 after surgery: 21.7% (10/46) vs. 40.9% (18/44), $P=0.043$]. However, no significant difference was observed on day 5 and 7 after surgery (day 5 after surgery: 15.2% (7/46) vs. 27.3% (12/46), $P=0.221$; day 7 after surgery: 10.9% (5/46) vs. 22.7% (10/44), $P=0.191$, Figure 2).

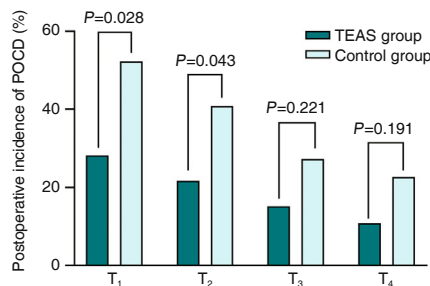


Figure 2. Postoperative Incidence of POCD in Two Groups ($\bar{x} \pm s$)

Notes: POCD: postoperative cognitive dysfunction; TEAS: transcutaneous electrical acupoint stimulation. T₁: day 1 post-surgery; T₂: day 3 post-surgery; T₃: day 5 post-surgery; T₄: day 7 post-surgery; the same below

The MMSE and MoCA scores of both groups declined to varying degrees after surgery. However, the MMSE scores on day 1, 3, and 5 after surgery were lower in the control group than in the TEAS group (all $P<0.05$, Figure 3A), whereas the MoCA scores on day 1, 3, 5, and 7 after surgery in the TEAS group were higher (all $P<0.05$, Figure 3B).

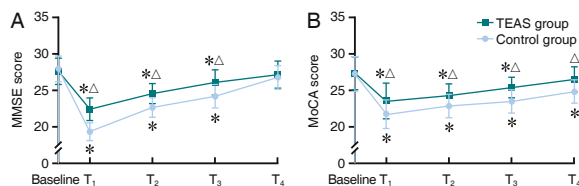


Figure 3. MMSE (A) and MoCA (B) Scores Measured at Different Time Points ($\bar{x} \pm s$)

Notes: MMSE: mini-mental status examination; MoCA: montreal cognitive assessment. * $P<0.05$ vs. baseline in the same group; $\Delta P<0.05$ vs. control group at the same time point

Plasma Levels of S100 β and NSE

Compared with baseline, the plasma levels of S100 β and NSE were significantly increased at 4, 8, 12, 24 h after surgery (all $P<0.05$). Compared with the control group, the plasma levels of S100 β and NSE were lower in the TEAS group at 4, 8, 12, 24 h after surgery (all $P<0.05$, Figures 4A and 4B).

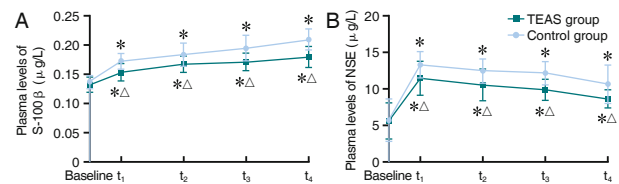


Figure 4. Plasma Levels of S100 β (A) and NSE (B) in Two Groups ($\bar{x} \pm s$)

Notes: t₁: 4 h post-surgery; t₂: 8 h post-surgery; t₃: 12 h post-surgery; t₄: 24 h post-surgery. * $P<0.05$ vs. baseline in the same group; $\Delta P<0.05$ vs. control group at the same time point

Safety

None of the participants showed signs of intense pain or communicated discomfort during the trial.

DISCUSSION

The results of the current study showed that electrical stimulation at the PC 6 and ST 36 acupoints reduced the incidence of POCD, improved postoperative MMSE and MoCA scores, and decreased plasma levels of S100 β protein and NSE. These findings indicated that TEAS is an effective complementary and alternative treatment to improve postoperative cognitive function in senior patients undergoing VATS.

The underlying pathophysiological mechanisms of POCD remain to be determined, although aging is the main reason for POCD.^(3,11) An increasing amount of evidence shows that the inflammatory response of brain tissue caused by surgical trauma is a key factor in the decline of neurocognitive function.⁽¹²⁾ Clinical trials have proved that electroacupuncture can inhibit inflammatory response, regulate the balance of pro-inflammatory and anti-inflammatory factors, has obvious curative effect in brain protection, and can effectively prevent or reduce the occurrence of POCD in senior patients after surgery.^(13,14) TEAS, a combination of transcutaneous electrical nerve stimulation and acupuncture points, is also an approach for inputting a specific low-frequency pulsed current into the body through the skin. TEAS overcomes some of the shortcomings of electroacupuncture, e.g., it is a noninvasive and relatively simple, is easily accepted by patients, and has no adverse reactions.⁽¹⁵⁾ There is evidence that TEAS and electroacupuncture have similar regulatory effects on the same brain functional area, and there is no significant difference between TEAS and acupuncture performed at the same acupoint.⁽¹⁶⁾

Acupuncture at ST 36 can activate most areas of the temporal lobe, and it is widely used in central nervous system diseases, such as motor sensory

disorders and cognitive dysfunction.⁽¹⁷⁾ PC 6 is included in the Hand-Jueyin Pericardium Meridian, and modern Chinese medicine theory believes that this acupoint is related to mental activity and thinking-related consciousness and is closely related to brain function. Therefore, PC 6 is the first choice for treating brain injury.⁽¹⁸⁾ Acupuncture at ST 36 and PC 6 can improve the blood supply of brain tissues, maintain the balance of oxygen and blood supply to the brain, and has a significant role in brain protection.⁽¹³⁾ The application of TEAS to these two acupoints has been shown to have some beneficial effects in brain-related disorders.

In our study, compared with the baseline value on day 1 before surgery, the MMSE and MoCA scores decreased to various extents. The MMSE scores on day 1, 3, and 5 after operation and MoCA scores on day 1, 3, 5, and 7 after operation in the TEAS group were higher than those in the control group. These results indicate that TEAS has a protective effect on POCD in senior patients after VATS lobectomy, reduced POCD at an early stage and improved postoperative cognitive function.

The International POCD Research Collaborative Group (ISPOCD), composed of 13 medical centers in 8 countries, collected 1,218 senior patients (>60 years old) who underwent elective non-cardiac surgery and found that 25.8% of patients developed POCD 1 week after surgery.⁽¹⁹⁾ In this study, the incidence of POCD in the TEAS and control groups on day 7 after surgery was 22.7% and 10.9%, respectively, and mainly occurred in the first 5 days after surgery. Moreover, the incidence of POCD on day 1 and 3 after operation in the TEAS group was lower than that in the control group; however, no significant difference was observed on day 5 and 7 after operation. In the ISPOCD study, old age, long anesthesia time, low education level, secondary operation, postoperative infection and respiratory complications were found to be risk factors for early POCD. We enrolled participants with no significant differences in age, height, body mass, anesthesia mode, anesthetic drugs, and operation time to eliminate the influence of other clinical factors as far as possible. Patients with other preoperative systemic diseases during the surgery were excluded.

Neuroinflammation can cause dysfunction of nerve cells and even death, leading to increased levels of biochemical markers of brain damage in the blood.

S100 β protein and NSE are considered potential markers of neuronal injury.⁽²⁰⁾ S100 β protein is an important protein in the nervous system, and it provides nutritional support to neuroepithelial cells and neurons. NSE is a key enzyme in the glycolysis pathway, which mainly exists in tissue and neuroendocrine tissues. When injured, considerable levels of S100 β protein and NSE leak into the cerebrospinal fluid and enter the bloodstream via the damaged blood-brain barrier. Therefore, the detection of their serum levels can be used as a diagnostic indicator of central nervous system (CNS) damage.^(21,22) A previous prospective study evaluated the association between POCD and serum S100 β and NSE levels.⁽²³⁾ The outcomes indicated that the serum levels of S100 β protein and NSE increased in senior patients after coronary artery bypass grafting, including immediately after the surgery; these findings suggest that surgical and anesthetic trauma are associated with neural damage that correlates with the incidence of POCD.

The results also showed that the serum S100 β and NSE levels of the two groups increased at various time points after surgery compared with those before surgery, indicating the participants experienced a certain level of CNS damage after surgery. However, the plasma levels of S100 β protein and NSE were lower in the TEAS group than in the control group at 4, 8, 12, and 24 h after surgery, indicating that TEAS was more effective than the control treatment in inhibiting CNS damage. Given the vulnerability of the nervous system in senior patients, TEAS reduces the damage to the CNS caused by surgery and anesthesia, further accelerates the repair of cognitive function, and reduces the incidence of POCD.

There are several limitations of this study that warrant discussion. First, the number of patients is relatively small. Second, we only studied elderly patients undergoing VATS. So the effects of preconditioning of TEAS on POCD in patients of other age groups or receiving other surgeries need further study. Third, we did not follow the long-term incidence of POCD. In fact, POCD may affect patient's cognitive function and quality of life for a long period after surgery, or even increase postoperative mortality.

Our results showed that TEAS could decrease the incidence of POCD in senior patients after VATS and enhance postoperative cognitive function. However, due to the lack of relevant studies, more detailed researches should be conducted to justify the effects of

TEAS in reducing the incidence of POCD.

Conflict of Interest

The authors declare that they have no competing interests.

Author Contributions

Wei H and Gan JH contributed equally to this work. Wei H and Gan JH designed the trial. Wei H and Huang JS drafted the manuscript. Zhao F, Xie ZQ and Xia ZY collected, analyzed and interpreted the data. All authors read and approved the final manuscript.

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